Thara Sivanandam

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PROFESSIONAL SUMMARY

Cybersecurity graduate student focused on cloud infrastructure, security engineering, and automation. Through AWS labs, network security research, Python development, and cloud-native projects, I have learned to approach technical problems by first identifying the operational or security gap, then building repeatable solutions, validating them through testing and monitoring, and documenting the process for future use. My experience includes deploying AWS environments, applying IAM least-privilege principles, troubleshooting Linux-based network labs, automating Python workflows, and designing secure cloud architectures. I am seeking an entry-level cloud, cybersecurity, DevSecOps, or IT operations role where I can contribute hands-on technical skills while continuing to grow in real-world production environments.

TECHNICAL SKILLS

Cloud Infrastructure: AWS (EC2, S3, IAM, VPC, Lambda, CloudWatch, Cognito, API Gateway, DynamoDB), Terraform, CloudFormation, Windows Server, Ubuntu Server, IPSec/IKEv2, Windows Defender

Programming and Automation: Python, Bash, Java, REST APIs, Flask, FastAPI

DevOps and CI/CD: GitHub, GitHub Actions, Linux, CI/CD pipelines, Infrastructure as Code (IaC)

Security: Cloud security, IAM least-privilege, encryption-at-rest, Wireshark, Burp Suite, Qualys, Scapy, Autopsy

Competencies: Cloud architecture design, cost optimisation, technical documentation, root cause analysis, remote collaboration, troubleshooting

EXPERIENCE

AWS Academy Cloud Foundations | Drexel University, Philadelphia, PA — *Jan 2026 – Apr 2026*

- Built and configured 10+ AWS lab environments using EC2, S3, IAM, VPC, Lambda, API Gateway, DynamoDB, and CloudWatch to develop hands-on cloud deployment and troubleshooting experience.
- Applied IAM least-privilege access, S3 bucket policies, subnet routing, and CloudWatch monitoring to understand how cloud teams secure, observe, and operate AWS environments.
- Evaluated cloud design decisions using the AWS Well-Architected Framework, comparing cost, reliability, security, and operational trade-offs before finalizing lab architectures.
- Learned how serverless patterns reduce infrastructure overhead by deploying Lambda functions triggered through S3 and API Gateway.

Network Security and Digital Forensics Lab Contribution | Drexel University, Philadelphia, PA — *May 2025 – Aug 2025*

- Configured and maintained 5 isolated Linux-based network lab environments for packet analysis, attack simulation, and digital forensic testing.
- Designed ARP poisoning and Man-in-the-Middle attack simulations using Wireshark and Scapy, then validated 3 detection countermeasures through packet capture analysis.
- Created shared configuration templates for a 4-person cohort to reduce setup inconsistencies and improve repeatability across lab sessions.
- Authored a technical report documenting attack procedures, detection methods, and remediation steps that became a shared reference for the cohort.

Python Developer Intern | Global Techno Solutions, Chennai, India *Jun 2021 – Aug 2021*

- Refactored 5 legacy Python modules used across 3 downstream data pipelines, improving script reliability and reducing workflow bottlenecks.
- Built a reusable logging and exception-handling library to standardize debugging across shared Python scripts.
- Automated 3 recurring data-processing workflows, reducing manual intervention and allowing the team to focus on higher-priority development work.
- Presented code updates during weekly reviews and contributed as an individual developer within an 8-person remote team.

PROJECTS

SafePlay: Child Safety and Parental Monitoring Platform *Dec 2025 – Present*

- Designed a secure AWS cloud architecture for a child safety platform using Cognito, Lambda, API Gateway, DynamoDB, S3, CloudWatch, and IAM.
- Built the security model around least-privilege IAM roles, encrypted storage, HTTPS-only API access, and OAuth2 authentication flows.
- Defined a GitHub Actions CI/CD workflow for automated testing, staged deployment, and DevSecOps-aligned delivery.
- Created architecture diagrams, API specifications, data flow maps, and a phased roadmap to make the system understandable and implementation ready.

Sophia: Personal Virtual Assistant *Nov 2024 - Mar 2025*

- Developed a modular Python virtual assistant with 5 components: application control, reminders, to-do management, NLP-based command parsing, and health symptom logging.
- Implemented structured logging and exception handling across all modules to improve debugging and runtime traceability.
- Conducted iterative testing and debugging to resolve edge cases in NLP parsing and scheduling conflicts.
- Learned how component-based design improves maintainability and makes future feature expansion easier.

EDUCATION

Master of Science in Cybersecurity | GPA: 3.89 / 4.0 *Mar 2025 – Sep 2026*
Drexel University, Philadelphia, PA | Coursework: Cloud Security, Network Security, Cryptography, Digital Forensics, Infrastructure Management

B.E. in Computer Science and Engineering | GPA: 3.43 / 4.0 *Oct 2020 – Jun 2024*
Velammal Institute of Technology, Tamil Nadu, India

CERTIFICATIONS

AWS Certified Cloud Practitioner - exam in active preparation *In-preparation*

Google Cybersecurity Professional Certificate - SIEM, incident response, Linux, network and cloud security *Nov 2024*

ADDITIONAL INFORMATION

- Authorised for full-time U.S. employment under F-1 OPT. No sponsorship required during OPT period.
- I am seeking an entry-level opportunity in roles like Cloud Engineer, AWS Support Engineer, DevSecOps Engineer, Infrastructure Operations, Cloud Security Analyst, IT Support.
- I am interested in Cloud cost optimisation, serverless architectures, quantum computing, CTF and bug bounty competitions, cyber defence strategy and Innovating thinking.